

SOLVED PRACTICE WORKBOOK

Algebra-Inequalities-and-Data-Sufficiency - CSAT Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Algebra-Inequalities-and-Data-Sufficiency - Solved Practice Workbook

Identity: csat-05: learner-v2:g1 | Approval: false

BASIC MCQS / REMEDIATION

Diagnostic set - Questions 1-16

Q1. Practice variant 1: Solve $3x + 2 = 11$.

A. 3 B. 4 C. 2 D. 5

Q2. Practice variant 1: Two numbers have sum 20 and difference 4. What is the larger number?

A. 13 B. 12 C. 11 D. 14

Q3. Practice variant 1: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q4. Practice variant 1: How many real solutions does $|x| = 4$ have?

A. 0 B. 1 C. 4 D. 2

Q5. Practice variant 1: Compare Quantity I: 3^2 and Quantity II: 8.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q6. Practice variant 1: What is x ? Statement 1: $x + 2 = 12$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q7. Practice variant 1: What is x ? Statement 1: $x^2 = 9$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Q8. Practice variant 1: For $0 \leq y \leq 5$, what is the maximum of $y(10-y)$?

A. 26 B. 24 C. 27 D. 25

Q9. Practice variant 2: Solve $3x + 3 = 15$.

A. 4 B. 5 C. 3 D. 6

Q10. Practice variant 2: Two numbers have sum 22 and difference 4. What is the larger number?

A. 14 B. 13 C. 12 D. 15

Q11. Practice variant 2: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q12. Practice variant 2: How many real solutions does $|x| = 5$ have?

A. 0 B. 1 C. 4 D. 2

Q13. Practice variant 2: Compare Quantity I: 4^2 and Quantity II: 15.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q14. Practice variant 2: What is x ? Statement 1: $x + 2 = 13$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q15. Practice variant 2: What is x ? Statement 1: $x^2 = 16$. Statement 2: $x > 0$.

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Q16. Practice variant 2: For $0 \leq y \leq 6$, what is the maximum of $y(12-y)$?

A. 37 B. 35 C. 38 D. 36

Graded set - Questions 17-32

Q17. Practice variant 3: Solve $3x + 4 = 19$.

A. 5 B. 6 C. 4 D. 7

Q18. Practice variant 3: Two numbers have sum 24 and difference 4. What is the larger number?

A. 15 B. 14 C. 13 D. 16

Q19. Practice variant 3: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q20. Practice variant 3: How many real solutions does $|x| = 6$ have?

A. 0 B. 1 C. 4 D. 2

Q21. Practice variant 3: Compare Quantity I: 5^2 and Quantity II: 24.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q22. Practice variant 3: What is x ? Statement 1: $x + 2 = 14$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q23. Practice variant 3: What is x ? Statement 1: $x^2 = 25$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Q24. Practice variant 3: For $0 \leq y \leq 7$, what is the maximum of $y(14-y)$?

A. 50 B. 48 C. 51 D. 49

Q25. Practice variant 4: Solve $3x + 5 = 23$.

A. 6 B. 7 C. 5 D. 8

Q26. Practice variant 4: Two numbers have sum 26 and difference 4. What is the larger number?

A. 16 B. 15 C. 14 D. 17

Q27. Practice variant 4: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q28. Practice variant 4: How many real solutions does $|x| = 7$ have?

A. 0 B. 1 C. 4 D. 2

Q29. Practice variant 4: Compare Quantity I: 6^2 and Quantity II: 35.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q30. Practice variant 4: What is x ? Statement 1: $x + 2 = 15$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q31. Practice variant 4: What is x ? Statement 1: $x^2 = 36$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Q32. Practice variant 4: For $0 \leq y \leq 8$, what is the maximum of $y(16-y)$?

A. 65 B. 63 C. 66 D. 64

Remedial and timed set - Questions 33-48

Q33. Practice variant 5: Solve $3x + 6 = 27$.

A. 7 B. 8 C. 6 D. 9

Q34. Practice variant 5: Two numbers have sum 28 and difference 4. What is the larger number?

A. 17 B. 16 C. 15 D. 18

Q35. Practice variant 5: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q36. Practice variant 5: How many real solutions does $|x| = 8$ have?

A. 0 B. 1 C. 4 D. 2

Q37. Practice variant 5: Compare Quantity I: 7^2 and Quantity II: 48.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q38. Practice variant 5: What is x ? Statement 1: $x + 2 = 16$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q39. Practice variant 5: What is x ? Statement 1: $x^2 = 49$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Q40. Practice variant 5: For $0 \leq y \leq 9$, what is the maximum of $y(18-y)$?

A. 82 B. 80 C. 83 D. 81

Q41. Practice variant 6: Solve $3x + 7 = 31$.

A. 8 B. 9 C. 7 D. 10

Q42. Practice variant 6: Two numbers have sum 30 and difference 4. What is the larger number?

A. 18 B. 17 C. 16 D. 19

Q43. Practice variant 6: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Q44. Practice variant 6: How many real solutions does $|x| = 9$ have?

A. 0 B. 1 C. 4 D. 2

Q45. Practice variant 6: Compare Quantity I: 8^2 and Quantity II: 63.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Q46. Practice variant 6: What is x ? Statement 1: $x + 2 = 17$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Q47. Practice variant 6: What is x ? Statement 1: $x^2 = 64$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Q48. Practice variant 6: For $0 \leq y \leq 10$, what is the maximum of $y(20-y)$?

A. 101 B. 99 C. 102 D. 100

PYQS AND ANSWER PRACTICE

Complete step-by-step solutions

Q1. Practice variant 1: Solve $3x + 2 = 11$.

A. 3 B. 4 C. 2 D. 5

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q2. Practice variant 1: Two numbers have sum 20 and difference 4. What is the larger number?

A. 13 B. 12 C. 11 D. 14

Correct answer: B. Adding the equations gives twice the larger number.

Q3. Practice variant 1: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q4. Practice variant 1: How many real solutions does $|x| = 4$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q5. Practice variant 1: Compare Quantity I: 3^2 and Quantity II: 8.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q6. Practice variant 1: What is x ? Statement 1: $x + 2 = 12$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q7. Practice variant 1: What is x ? Statement 1: $x^2 = 9$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q8. Practice variant 1: For $0 \leq y \leq 5$, what is the maximum of $y(10-y)$?

A. 26 B. 24 C. 27 D. 25

Correct answer: D. Complete the square: $y(2n-y)=n^2-(y-n)^2$, maximised at $y=n$.

Q9. Practice variant 2: Solve $3x + 3 = 15$.

A. 4 B. 5 C. 3 D. 6

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q10. Practice variant 2: Two numbers have sum 22 and difference 4. What is the larger number?

A. 14 B. 13 C. 12 D. 15

Correct answer: B. Adding the equations gives twice the larger number.

Q11. Practice variant 2: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q12. Practice variant 2: How many real solutions does $|x| = 5$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q13. Practice variant 2: Compare Quantity I: 4^2 and Quantity II: 15.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q14. Practice variant 2: What is x ? Statement 1: $x + 2 = 13$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q15. Practice variant 2: What is x ? Statement 1: $x^2 = 16$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q16. Practice variant 2: For $0 \leq y \leq 6$, what is the maximum of $y(12-y)$?

A. 37 B. 35 C. 38 D. 36

Correct answer: D. Complete the square: $y(2n-y) = n^2 - (y-n)^2$, maximised at $y=n$.

Q17. Practice variant 3: Solve $3x + 4 = 19$.

A. 5 B. 6 C. 4 D. 7

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q18. Practice variant 3: Two numbers have sum 24 and difference 4. What is the larger number?

A. 15 B. 14 C. 13 D. 16

Correct answer: B. Adding the equations gives twice the larger number.

Q19. Practice variant 3: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q20. Practice variant 3: How many real solutions does $|x| = 6$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q21. Practice variant 3: Compare Quantity I: 5^2 and Quantity II: 24.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q22. Practice variant 3: What is x ? Statement 1: $x + 2 = 14$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q23. Practice variant 3: What is x ? Statement 1: $x^2 = 25$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q24. Practice variant 3: For $0 \leq y \leq 7$, what is the maximum of $y(14-y)$?

A. 50 B. 48 C. 51 D. 49

Correct answer: D. Complete the square: $y(2n-y) = n^2 - (y-n)^2$, maximised at $y=n$.

Q25. Practice variant 4: Solve $3x + 5 = 23$.

A. 6 B. 7 C. 5 D. 8

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q26. Practice variant 4: Two numbers have sum 26 and difference 4. What is the larger number?

A. 16 B. 15 C. 14 D. 17

Correct answer: B. Adding the equations gives twice the larger number.

Q27. Practice variant 4: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q28. Practice variant 4: How many real solutions does $|x| = 7$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q29. Practice variant 4: Compare Quantity I: 6^2 and Quantity II: 35.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q30. Practice variant 4: What is x ? Statement 1: $x + 2 = 15$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q31. Practice variant 4: What is x ? Statement 1: $x^2 = 36$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q32. Practice variant 4: For $0 \leq y \leq 8$, what is the maximum of $y(16-y)$?

A. 65 B. 63 C. 66 D. 64

Correct answer: D. Complete the square: $y(2n-y) = n^2 - (y-n)^2$, maximised at $y=n$.

Q33. Practice variant 5: Solve $3x + 6 = 27$.

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q34. Practice variant 5: Two numbers have sum 28 and difference 4. What is the larger number?

A. 17 B. 16 C. 15 D. 18

Correct answer: B. Adding the equations gives twice the larger number.

Q35. Practice variant 5: If $-2x > 8$, which statement is correct?

A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$

Correct answer: C. Dividing an inequality by a negative reverses its sign.

Q36. Practice variant 5: How many real solutions does $|x| = 8$ have?

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.

Q37. Practice variant 5: Compare Quantity I: 7^2 and Quantity II: 48.

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.

Q38. Practice variant 5: What is x ? Statement 1: $x + 2 = 16$. Statement 2: x is positive.

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.

Q39. Practice variant 5: What is x ? Statement 1: $x^2 = 49$. Statement 2: $x > 0$.

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.

Q40. Practice variant 5: For $0 \leq y \leq 9$, what is the maximum of $y(18-y)$?

A. 82 B. 80 C. 83 D. 81

Correct answer: D. Complete the square: $y(2n-y)=n^2-(y-n)^2$, maximised at $y=n$.

Q41. Practice variant 6: Solve $3x + 7 = 31$.

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Subtract the constant, divide by 3, then substitute back.

Q42. Practice variant 6: Two numbers have sum 30 and difference 4. What is the larger number?

A. 18 B. 17 C. 16 D. 19

Correct answer: B. Adding the equations gives twice the larger number.

Q43. Practice variant 6: If $-2x > 8$, which statement is correct?A. $x > -4$ B. $x < 4$ C. $x < -4$ D. $x > 4$ **Correct answer: C.** Dividing an inequality by a negative reverses its sign.**Q44. Practice variant 6: How many real solutions does $|x| = 9$ have?**

A. 0 B. 1 C. 4 D. 2

Correct answer: D. For a positive a , $|x|=a$ gives $x=a$ and $x=-a$.**Q45. Practice variant 6: Compare Quantity I: 8^2 and Quantity II: 63.**

A. Quantity I is greater B. Quantity II is greater C. They are equal D. Cannot be determined

Correct answer: A. Their difference is exactly 1.**Q46. Practice variant 6: What is x ? Statement 1: $x + 2 = 17$. Statement 2: x is positive.**

A. Statement 2 alone is sufficient B. Statement 1 alone is sufficient C. Both together are required D. Even both are insufficient

Correct answer: B. Statement 1 fixes x uniquely; positivity alone does not.**Q47. Practice variant 6: What is x ? Statement 1: $x^2 = 64$. Statement 2: $x > 0$.**

A. Statement 1 alone is sufficient B. Statement 2 alone is sufficient C. Both statements together are sufficient D. Even both are insufficient

Correct answer: C. Statement 1 leaves plus/minus values; Statement 2 selects the positive one.**Q48. Practice variant 6: For $0 \leq y \leq 10$, what is the maximum of $y(20-y)$?**

A. 101 B. 99 C. 102 D. 100

Correct answer: D. Complete the square: $y(20-y)=n^2-(y-n)^2$, maximised at $y=n$.**Verified PYQ routing ledger**

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	35	Reversed-digit equation	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	50	Linear-cost validation	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	59	Data sufficiency: integers	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	60	Data sufficiency: transfer rate	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	64	Data sufficiency: integer pair	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	65	Data sufficiency: shares	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	66	Data sufficiency: class size	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	67	Data sufficiency: prime triple	A (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	68	Data sufficiency: integrality	D (supplied)	Test each statement alone, then together, seeking uniqueness.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	69	Data sufficiency: article prices	C (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	70	Data sufficiency: score comparison	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	73	Data sufficiency: reversed ages	A (supplied)	Test each statement alone, then together, seeking uniqueness.
2024	76	Coded inequalities	D (supplied)	Translate signs and test the full allowed range.
2025	19	Chained-inequality validation	D (supplied)	Translate signs and test the full allowed range.
2025	26	Bounded-expression ratio	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	68	Data sufficiency: factor count	D (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	69	Data sufficiency: digit product	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	70	Data sufficiency: family relation	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	73	Data sufficiency: match outcome	D (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	74	Data sufficiency: positivity	B (supplied)	Test each statement alone, then together, seeking uniqueness.
2025	75	Quantitative comparison: ratio	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	76	Quantitative comparison: product	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	77	Quantitative comparison: set	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	79	Unit-interval inequalities	A (supplied)	Translate signs and test the full allowed range.
2026	1	Data sufficiency: coin denomination	A (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	2	Data sufficiency: compare reals	D (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	3	Data sufficiency: parity	C (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	4	Data sufficiency: set-membership logic	D (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	5	Data sufficiency: prime determination	A (provisional)	Test each statement alone, then together, seeking uniqueness.
2026	8	Weight equations comparison	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	21	Inequality-range statements	B (provisional)	Translate signs and test the full allowed range.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2026	38	Linear marking-scheme equations	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	80	Rate-constraint optimisation	D (provisional)	Use the total crossing length.

Final error-log checklist

- Was the family classified correctly?
- Were units, domains, quantifiers and constraints copied correctly?
- Was the first-principles relation written before a shortcut?
- Was the selected option independently recomputed or supported by the passage?
- Was the mistake logged and followed by a fresh remedial item?